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Tick size, NYSE rule 118, and ex-dividend day stock price behavior[☆]

Keith Jakob^a, Tongshu Ma^{b,*}

^a*School of Business Administration, University of Montana, Missoula, MT 59812, USA*

^b*David Eccles School of Business, University of Utah, Salt Lake City, UT 84112, USA*

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Abstract

Bali and Hite (1998) and Dubofsky (1992) propose models in which market microstructure effects play a role in the ex-dividend day price drop anomaly. Bali and Hite suggest that the anomaly is caused solely by price discreteness, while Dubofsky suggests that NYSE Rule 118 is also involved. We test these models by examining cum- to ex-day price drops during the one-eighth, one-sixteenth, and decimal tick size regimes. While the evidence is qualitatively consistent with Dubofsky's predictions, neither model is satisfactory in a quantitative sense. One of our main empirical findings is that no significant decline was evident in the magnitude of the ex-day anomaly after the tick size reduction.

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1. Introduction

According to Miller and Modigliani (1961), in a frictionless perfect capital market with no uncertainty, the share price following a dividend should fall by exactly the amount of the dividend paid on each share. However, empirical works on this issue

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*Corresponding author. Tel.: +1-801-581-7506; fax: +1-801-581-7214.

E-mail address: fintsm@business.utah.edu (T. Ma).

have consistently found that, on average, stock prices drop by less than the dividend amount on the ex-dividend date. This unusual ex-dividend day price behavior has many economists searching for an acceptable explanation.

In a widely cited study, [Elton and Gruber \(1970\)](#) note that because investors care only about after-tax returns, a higher tax on dividend income than on capital gains will result in an ex-dividend price drop that is smaller than the dividend. They find that the price drop to dividend ratio is only 0.778 for a large sample of dividend distributions in 1966 and 1967. They also find that the price drop to dividend ratio is directly related to the level of the dividend yield. Elton and Gruber believe that this relation stems from the formation of tax clienteles for stocks of different dividend yields. They interpret the empirical findings as evidence consistent with the hypothesis that investors have a tax-induced preference for capital gains over dividends.

Both [Kalay \(1982\)](#) and [Miller and Scholes \(1982\)](#) suggest that the Elton and Gruber tax clientele hypothesis appears to be inconsistent with a no-arbitrage equilibrium. Another prominent study, by [Frank and Jagannathan \(1998\)](#), casts serious doubt on the validity of the tax-effect explanation. Frank and Jagannathan find that in Hong Kong the average stock price drop on ex-dividend days is also smaller than the dividend amount, even though neither dividends nor capital gains are taxed.

Some recent work on the ex-day price drop phenomenon focuses on market microstructure effects as potential explanations for the observed anomaly. Two important papers are found in this literature: [Dubofsky \(1992\)](#) and [Bali and Hite \(1998\)](#). Dubofsky argues that ex-dividend day excess returns arise from the mechanics of NYSE Rule 118, AMEX Rule 132, and the fact that prices are constrained to discrete tick multiples (i.e., price discreteness). Bali and Hite suggest that the ex-day price drop anomaly is caused solely by price discreteness.

NYSE Rule 118 and AMEX Rule 132 dictate that, on ex-cash dividend days, open limit orders to buy stocks are reduced by the cash dividend amount. With discrete prices, if the resulting price is not a tick multiple, it is further lowered to the next tick. Prices in limit sell orders are not changed. For example, if the tick size is \$0.125 (\$1/8) and a dividend is \$0.15, then the price of limit buy orders will be adjusted down by \$0.25 and limit sell orders will not be adjusted. Dubofsky argues that if on the ex-day the bid-ask spread is still constrained by the adjusted open limit orders, then the bid-ask spread is wider than normal and not symmetric around the expected ex-dividend day adjusted price. This asymmetry will lead to smaller measured price drops than dividends.

[Bali and Hite \(1998\)](#) present an ex-day price drop model in which prices are restricted to multiples of a tick. Their model indicates that investors will never be willing to pay more for a dividend than its value, and they show that the equilibrium ex-day price drop will be the amount of the dividend rounded to the next smaller tick. For example, if the tick size is \$1/8 and the dividends are \$0.20 and \$0.25, then the ex-day price drops will both be \$0.125.

In this paper we conduct direct empirical tests of the impact of price discreteness and NYSE Rule 118 on ex-dividend day price behavior. We try to determine whether

the data are consistent with Dubofsky's model or that proposed by Bali and Hite. Recently, the NYSE converted from \$1/8 ticks to \$1/16 ticks and finally to decimals. We use data in all three tick size regimes to test the two models. We report five main findings. (1) For the most common dividend amounts, the ex-day price drop is just as likely to be the tick above the dividend as to be the tick below the dividend. (2) Regression analysis shows that the average price drop does not equal the tick below the dividend. (3) For all three tick size regimes, the average cum-day bid to ex-day bid price drop is larger than the dividend on average, which is larger than the cum-day ask to ex-day ask price drop. (4) The opening bid-ask spread is larger on the ex-day than on the cum-day. (5) No significant reduction is found in the difference between the dividend amount and average ex-day price drop as the tick size is reduced. All of these findings are inconsistent with Bali and Hite (1998). While these results are qualitatively consistent with Dubofsky (1992), they contradict Dubofsky at a quantitative level if we take Dubofsky's fully constrained model literally.¹ Therefore, something important is still missing from these models.

Beyond our general findings, two features of our study are worth mentioning. First, we use a relatively large sample to perform an extensive examination and comparison of the different price drop relations using cum-day close to ex-day open prices as well as cum-day close to ex-day close prices. Past research on ex-dividend day price behavior mostly examines cum-day close to ex-day close price drops.² A major difficulty in the empirical research is that the close-to-close price drops are very noisy. Eades et al. (1994) document large variations over time in average ex-dividend day abnormal returns. Boyd and Jagannathan (1994) find that the ex-day price drop is very noisy, and hence inference based on one or a few years' data will be imprecise. We address this problem by using a relatively large sample to examine the cum-day close to ex-day open price drop in addition to the close-to-close price drop. This allows us to make more precise inferences because the overnight price drop is much less noisy than the close-to-close price drop. For example, in our sample, the overnight price drop has a standard deviation of \$0.399, whereas the cum-day close to ex-day close price drop has a standard deviation of \$0.728. This means that, in terms of inference about the average ex-day price drop, one observation in our sample is equivalent to nearly two (i.e., $0.728/0.399 = 1.82$) observations in a sample that contains only close prices. Our sample, therefore, of more than 52,000 observations is equivalent to more than 94,000 observations of close-to-close price drops.

¹ Even Dubofsky advises not taking his fully constrained model literally. But without doing so, the model gives no quantitative prediction on the amount of the ex-day price drop.

² There are a few exceptions. Eades et al. (1984) studied cum-day close to ex-day open price drops for securities in the Dow Jones 30 from July 1962 to June 1972; Lakonishok and Vermaelen (1986) studied overnight price drops for NYSE and AMEX securities from January 1970 to December 1981. Dubofsky (1992) also studied overnight price drops for a subsample of NYSE stocks from August 1962 to September 1972. Our contribution is that our use of the overnight price drop is partly motivated by the fact that the overnight price drops are less noisy, and price noisiness is a major obstacle in the empirical study of ex-day price drops, as pointed out by Eades et al. (1994) and Boyd and Jagannathan (1994).

Second, we use a relatively large sample to examine ex-day drops in quoted prices and changes in the bid-ask spread. Past research mainly uses transaction prices. Using the bid-ask spread and quote data, we can test the model predictions about changes in market quotes. For example, our finding that the price drop from cum-day close ask to ex-day open ask is substantially smaller than that from cum-day close bid to ex-day open bid is consistent with Dubofsky's (1992) predictions.

Recently, Graham et al. (2003) also examine the effect of tick size reduction on the ex-dividend day price drop anomaly. Like us, they find that after the decimalization the ex-dividend day price drop anomaly did not go away. But they attribute the persistence of the anomaly to a tax change around the time of the decimalization and conclude that their results support the tax-effect explanation of the ex-day price drop anomaly.

The remainder of the paper is organized as follows. In Section 2 we present the testable implications of the Dubofsky and Bali and Hite models. In Section 3 we discuss the data and econometric modeling used in the analysis. In Section 4 we present our major empirical findings. Section 5 concludes the paper. We give a detailed discussion of some econometric issues in the Appendix.

2. Model predictions and testable implications

Bali and Hite (1998) and Dubofsky (1992) make different predictions about the size of ex-day price drops. Both models assume that no informational event occurs from the cum- to the ex-day. Bali and Hite's model predicts that the ex-day price drop will always equal the tick below the dividend. That is, when the tick size is \$0.125, dividends less than or equal to \$0.125 should have price drops of zero; dividends greater than \$0.125 but less than or equal to \$0.25 should have price drops of \$0.125; and so on. If we use $\underline{D}$ to denote the tick below the dividend, then Bali and Hite predict that the price drop is exactly

$$\Delta P = \underline{D}. \quad (1)$$

Bali and Hite's model has four important implications. (1) If the frequency distribution of the ex-day price drops is examined, then cases in which price drops are equal to the tick below the dividend should be more plentiful than cases in which price drops are equal to the tick above the dividend. This should be true even if informational events occasionally occur on ex-days. (2) The average price drop should never exceed the dividend amount for any dividend size, and for dividends less than or equal to the prevailing tick size, price drops should equal zero. (3) Asymmetric changes should not be expected in the bid and ask prices from cum- to ex-day. (4) Because $(D - \underline{D})$ is always less than the tick size, as the tick size is reduced, the magnitude of the ex-day price drop anomaly should decline, and with decimalization the anomaly should essentially disappear.

The key feature of Dubofsky's (1992) model is the exchange mandated price adjustment for existing limit orders on the ex-dividend day. According to NYSE Rule 118 and AMEX Rule 132, on the ex-day existing limit buy orders are reduced

by the dividend amount. If the resulting price is not a multiple of a tick, the limit buy price is further reduced to the next tick. The existing limit sell orders are not adjusted. Dubofsky's model gives an exact prediction of the ex-day price drop only under the following restrictive assumptions.

Assumption A. *On the ex-day, bid and ask prices are fully constrained to the adjusted limit order prices.*

Assumption B. *On both the cum- and the ex-day, trades occur on the bid and ask sides of the quotes with an equal probability of 50%.*

If $\bar{D}$ is used to denote the tick multiple that is greater than or equal to the dividend amount, ΔP_a the drop in ask price from cum- to ex-day, ΔP_b the drop in bid price, and ΔP the drop in transaction price, then, under Assumptions A and B, Dubofsky's model makes the following predictions about ex-day price drops:

$$\Delta P_a = 0, \quad (2)$$

$$\Delta P_b = \bar{D} \quad (3)$$

and

$$\Delta P = (\Delta P_a + \Delta P_b)/2 = \bar{D}/2. \quad (4)$$

Under Assumptions A and B, Dubofsky's four predictions are dramatically different from those of [Bali and Hite \(1998\)](#): (1) In many cases ex-day price drops will be $\bar{D}$, according to Eq. (3). (2) For dividends less than or equal to the tick size, the average price drop will be greater than zero. For the dividends less than one-half of the tick size, the average ex-day price drop will be larger than the dividends. (3) From cum- to ex-day, the drop in quoted ask price will be smaller than the drop in quoted bid price, and the bid-ask spread will widen. (4) Reducing the tick size means $\bar{D}$ declines for any given dividend amount. According to Eq. (4), the average price drop will become smaller for a given dividend amount. As a result, reducing the tick size will exacerbate the ex-day price drop anomaly.

Dubofsky recognizes that Assumptions A and B will not be fully satisfied in most cases. Dubofsky suggests that for the smallest of dividends the prices will conform most closely to the assumptions. However, as dividend size increases, the outstanding limit sell orders, which are not adjusted according to the exchange rules, will more likely be replaced by newly submitted limit sell orders and specialist interventions. The limit sell orders will be added to the limit order book, and if they offer better prices (i.e., lower) than the current ask price, the quote will be adjusted. This action will violate Assumption A, because the quoted ask price will no longer be constrained to the unadjusted limit sell orders. Specialist interventions can occur in two important ways. First, specialists can behave exactly like a better limit sell order and adjust the quoted ask price lower (again violating Assumption A). Second, specialists can leave the limit order book unadjusted but willingly make trades at better prices than the quoted ask price. This behavior will violate Assumption B,

because trades will no longer be evenly distributed between the bid and the ask price. In all of the cases, the specialist and limit order activities will reduce the transaction price for buy-initiated transactions, and this will make ΔP greater than $\bar{D}/2$.

Even though the assumptions are likely to be violated, unless something happens on the ex-day that totally offsets the effect of the mechanical limit order adjustment, some effect of the adjustment will remain. Hence the model predictions we outlined above will still be valid (after straightforward modifications). Therefore, the model's assumptions do not have to hold exactly to give the above qualitative predictions. For example, because the exchange-mandated adjustment in the bid price of the existing limit orders is $\bar{D}$, so long as some trades occur on the bid side on both cum- and ex-days, we still expect to see many price drops equal to $\bar{D}$. So, prediction (1) still holds. Similarly, predictions (2) and (4) will also be true after we make the following minor alterations. Regarding prediction (2), for dividends less than or equal to the tick size, the average price drop could be greater than zero. For dividends less than one-half of the tick size, the average ex-day price drop could be larger than the dividends. Regarding prediction (4), it could be the case that smaller tick size does not imply a smaller ex-day price drop anomaly. As for prediction (3), because the exchange mandates that prices in the limit buy orders are reduced by $\bar{D}$ while the prices in the limit sell orders are not adjusted, as long as in some cases the adjusted prices constrain the ex-day quotes, we should see larger average drops in bid prices and smaller average drops in ask prices from cum- to ex-day. So, prediction (3) still holds.

2.1. Average price drop as a function of the dividend amount

Both Bali and Hite (1998) and Dubofsky (1992) predict a saw-toothed pattern regarding the relation between ex-day price drop and the dividend amount.³ In both cases the average ex-day price drop as a function of dividend size should experience a jump right after the dividend size becomes larger than a tick multiple. They both document evidence of this saw-toothed pattern in their respective papers.

However, the two models make different predictions about the exact shape of the saw-toothed pattern. Bali and Hite predict that, between ticks, the average price drop does not increase with dividend size [see Eq. (1)]. Dubofsky's price drop predictions depend on the amount of specialist interventions and new limit order submission. If, as Dubofsky argues, new limit orders and specialist intervention increase with dividend size, then, within ticks, the average price drop should increase with the dividend amount.

We can test these differences in the model predictions in the regression

$$\Delta P_{t,i} = a + b\bar{D}_{t,i} + c\delta_{t,i} + \varepsilon_{t,i}, \quad (5)$$

where $t = 1, 2, \dots, T$ are the ex-dividend days in the sample; for each t , $i = 1, 2, \dots, N_t$ are the stocks that go ex-dividend on date t . ΔP is the price drop from

³ For Bali and Hite, the saw-toothed pattern is implied by Eq. (1). For Dubofsky, the pattern is implied by Eq. (4). As we discussed in the text, the main implications of Eq. (4), and therefore, the implied saw-toothed pattern, will be present even if Assumptions A and B do not always hold exactly.

cum- to ex-day, $\underline{D}$ is the tick below the dividend, δ is $(D - \underline{D})$, and ε is the error term. With this regression specification Bali and Hite predict that $a = c = 0$ and $b = 1$. Dubofsky does not give an exact prediction about the ex-day price drop. However, if we assume that the ex-day price drop follows a linear relation as in Eq. (5), then Dubofsky (1992) implies that $b > 0$ and $c > 0$.

To summarize, Bali and Hite's and Dubofsky's model predictions differ in five measurable ways: the four aspects previously discussed and the different predictions about the coefficients in Eq. (5). Eq. (5) is a valid test of Bali and Hite's model. But it is a valid test of Dubofsky only under the additional assumption that it is the correct specification for the ex-day price drop. In Section 4 we will examine which of the predictions are consistent with the data.

We can interpret Dubofsky (1992) at two different levels depending on whether to assume that Assumptions A and B are fully satisfied. If Assumptions A and B are fully satisfied, then the model gives exact prediction on the ex-dividend day price drop as in Eqs. (2)–(4). These predictions are easily rejected. For example, the ex-day drop in ask price is clearly significantly positive, contradicting Eq. (2). Dubofsky (1992) advocates not taking his model literally. Then the model does not give an exact prediction about the price drops on ex-days. We feel that even in this case it is still valuable to examine whether Dubofsky gives the right qualitative prediction along different dimensions. However, giving only the right qualitative predictions cannot be considered a complete explanation of the ex-day price drop anomaly.

3. Data description and econometric modeling

Our data come from the Center for Research in Security Prices (CRSP) database and the NYSE Trade and Quote (TAQ) database along with the list of decimal conversion dates available from the NYSE. We collect information about all the ordinary, cash, and taxable dividend distributions made to NYSE stocks from January 1993 to December 2001 from the CRSP database. We exclude cases in which a firm made multiple distributions on the same date. We collect information about dividend amount, prices, returns, and trading volumes from CRSP. We collect the market trade and quote data at market open and close from January 1993 to December 2001 from the TAQ database. We merge the two sources of information and use the merged sample in our empirical study. Our sample includes only those dividend distributions that have valid dividend amount, price, volume, and market quote data. The merged sample has 52,293 observations. To eliminate the effects of outliers, we require the cum-day close price to be at least \$2.00. This leaves 52,179 observations.

The NYSE reduced tick size from \$1/8 to \$1/16 on June 24, 1997.⁴ It further reduced tick size to one cent for three pilot groups of stocks in August, September,

⁴ Using TAQ, we find 122 NYSE stocks that were already trading in \$1/16 before June 24, 1997. We find two dividend distributions before June 24, 1997 when the stock is trading in \$1/16 tick size.

and December 2000, respectively, and all other stocks on January 29, 2001. We divide our sample into three subsamples according to the stock's tick size at the time of the dividend distribution. There are 24,270 observations when the tick size is \$1/8, 22,512 observations when the tick size is \$1/16, and 5,397 observations when the tick size is \$0.01.

Data in ex-dividend studies have an important feature that many different stocks may go ex-dividend on the same day. This is referred to as dividend clustering. Most previous studies do not control for this problem in their statistical inference. One exception is Hayashi and Jagannathan (1990), who use a panel data model to allow common shocks in the stock market to affect ex-day returns of stocks that go ex-dividend on the same day. We follow the same method to deal with the dividend clustering problem.

Consider testing the prediction that $\Delta P = \underline{D}$. Bali and Hite (1998) apply ordinary least squares (OLS) to Eq. (5). Because of dividend clustering, the error terms $\varepsilon_{t,i}$ and $\varepsilon_{t,j}$ are likely to be correlated. Hence the standard errors in the OLS regression are incorrect. We correct this econometric problem by using Hayashi and Jagannathan's (1990) panel data model.

Following Hayashi and Jagannathan (1990), we allow a fixed effect for each ex-dividend day in modeling the ex-day price drop:

$$\Delta P_{t,i} = a + \sum_{t=1} a_t (\text{fixed effect})_t + b \underline{D}_{t,i} + c \delta_{t,i} + \varepsilon_{t,i}. \quad (6)$$

This accommodates the situation in which a favorable shock moves prices of all stocks on a particular date, thus affecting ex-day price drops of all stocks that go ex-dividend on that date. We assume that the error terms in Eq. (6) have zero mean and the same variance and that they are pairwise uncorrelated.

We will also apply this fixed-effect panel data model to examine other quantities such as the difference between the dividend amount and price drop and changes in bid-ask spread from cum- to ex-day. In the first case, the model will be

$$D_{t,i} - \Delta P_{t,i} = a + \sum_{t=1} a_t (\text{fixed effect})_t + \varepsilon_{t,i}. \quad (7)$$

Then the statement that ex-day price drops on average are smaller than the dividend is the same as the hypothesis that $a > 0$ in Eq. (7). This hypothesis is easily tested.

To examine whether the bid-ask spread widens from the cum- to the ex-day, we again perform the regression on Eq. (7) with the dependent variable being $\Delta \text{Spread}_{t,i}$, and then test whether a is greater than zero.

We relegate the details of the fixed-effect panel data model to the Appendix.

4. Empirical results

We now discuss our empirical findings.

4.1. Frequency distribution of the ex-day price drops

For our first test we examine the frequency distribution of the ex-day price drop. [Bali and Hite \(1998\)](#) predict that many more price drops should be equal to $\underline{D}$ than equal to $\overline{D}$. However, according to [Dubofsky \(1992\)](#), many of the ex-day price drops should be $\overline{D}$. [Table 1](#) presents the frequency distributions for the \$1/8 tick size data. (We do not present results for the other tick size regimes because the data are too noisy.) The table presents the results for the \$0.05, \$0.10, \$0.15, \$0.20, and \$0.25 dividends, given that these are the most common dividend sizes in the data. For each dividend amount, there are almost equal numbers of price drops at $\overline{D}$ and at $\underline{D}$. This is also true for the other dividend sizes not presented in the table. For all of the dividends in the \$1/8 tick regime, the frequency distribution shows that 30.67% of the close-to-open price drops equal $\underline{D}$ and 30.85% of the price drops equal $\overline{D}$. For the close-to-close price drop, the frequencies are 20.6% and 20.1%, respectively. (These results are not reported in [Table 1](#).) So it appears equally likely that the ex-day price drop is $\underline{D}$ or $\overline{D}$.

4.2. Average price drops for small dividends

According to [Bali and Hite \(1998\)](#), the average price drop should be no greater than the dividend for any dividend amount, and for dividends less than or equal to

Table 1

Some frequency distributions of ex-day price drops

This table reports the frequencies of overnight price drops for dividends equal to \$0.05, \$0.10, \$0.15, \$0.20, and \$0.25, for the subsample of the data with the tick size of \$1/8 (January 1993 to June 1997). We include only ordinary, cash, and taxable dividend distributions made to NYSE stocks with cum-day prices of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Numbers of cases in which price drops equal to $\underline{D}$ or $\overline{D}$ are in bold font.

Price drops (in dollars)	Dividend amounts (in dollars)				
	0.05	0.1	0.15	0.2	0.25
−0.5	9	3	4	4	4
−0.375	9	8	3	4	3
−0.25	19	15	10	5	10
−0.125	108	96	25	22	17
0	492	242	157	78	109
0.125	423	272	221	119	148
0.25	147	139	220	124	176
0.375	37	41	88	62	96
0.5	12	12	40	18	41
0.625	5	4	2	5	9
All other	17	9	17	14	21
Number of observations	1,278	841	787	455	634
Average price drop	0.0739	0.0838	0.1828	0.1734	0.1914
$\Delta P/D$	1.478	0.838	1.219	0.867	0.766

Table 2

Average price drops from cum- to ex-day for small dividends

When tick size is \$1/8, the sample contains all dividends that are no greater than \$0.0625; when tick size is \$1/16, the sample contains all dividends that are no greater than \$0.03125; when tick size is \$0.01, the sample contains all dividends that are no greater than \$0.025. For each subsample, we first calculate the mean of (price drop – dividend) for each ex-dividend day over all stocks that become ex-dividend on that day. Then we calculate the mean and standard deviation (in parenthesis) of these daily means, in which the standard deviation is White's heteroskedasticity consistent one.

Tick size regime	Mean dividend	Mean of (price drop – dividend)	
		Price drop is measured from cum-close to ex-day open	Price drop is measured from cum-close to ex-day close
Tick = \$1/8 (No. obs = 953)	0.0422*** (0.0003)	0.03036*** (0.00375)	–0.00257 (0.00865)
Tick = \$1/16 (No. obs = 596)	0.0218*** (0.0002)	0.01942* (0.01129)	0.00198 (0.02963)
Tick = \$0.01 (No. obs = 144)	0.0175*** (0.0005)	–0.02791** (0.02202)	–0.03349 (0.04559)

* Significantly different from 0 at the 10% confidence level.

** Significantly different from 0 at the 5% confidence level.

*** Significantly different from 0 at the 1% confidence level.

the prevailing tick size, price drops should equal zero. According to Dubofsky (1992), for dividends that are smaller than half a tick, the average price drop could be larger than the dividend amount. In Table 1 the \$0.05 dividends are smaller than half of the \$0.125 tick size. For these cases, we report an average price drop of \$0.0739. This finding is consistent with Dubofsky. For a more detailed analysis we report the average price drops for all small dividends in Table 2. Here the hypothesis we are testing is whether the mean of (price drop – dividend) is positive or negative. According to Bali and Hite, it should be negative. But according to Dubofsky, it could be positive. Table 2 shows that if the ex-day price drop is measured from cum-day close to ex-day open, the average of (price drop – dividend) is significantly positive at 1% and 10% significant level, for the \$1/8 and \$1/16 tick size regime, respectively.⁵ Furthermore, no cases arise in which the average is significantly negative. This evidence again favors Dubofsky. (In the empirical analysis here and below, we assume an actual or effective tick size of \$0.05 for the decimalized subsample. This is because we find that people have a tendency to trade at five-cent intervals even though the nominal tick size is only one cent. We find that about 50%

⁵ Because of ex-day clustering, in arriving at Table 2, we first calculate the average of the price drop minus dividend for each day (across different distributions) when some stocks are distributing small dividends. Then we calculate the mean of these daily averages and their heteroskedasticity consistent standard error according to White (1980). We do not apply the panel data model here because the number of observations in each day is usually small, and hence the asymptotic results of the panel data model are unlikely to hold [see Eq. (A.6)]. Applying the panel data regression model yields qualitatively similar results.

of the trades occur at five-cent intervals during the decimalized period. We point out that this assumption is in neither Bali and Hite, 1998, nor Dubofsky, 1992. The true effective tick size of one cent instead of five cents yields qualitatively similar empirical results.)

4.3. Average price drop as a function of the dividend amount

For our third set of tests we begin with a replication of Bali and Hite's graphical analysis by plotting the average price drops for different dividend groups in Fig. 1. Panels A and B are for the subsample with the tick size of \$1/8, Panels C and D are for the subsample with the tick size of \$1/16, and Panels E and F are for the subsample with the tick size of \$0.01. Price drops in Panels A, C, and E are measured from close to close. Price drops in Panels B, D, and F are measured from cum-day close to ex-day open. Because our first subsample runs from 1993 to 1997, whereas Bali and Hite used data from 1962 to 1994, we can regard our first subsample as an out-of-sample check of their results. Following the method of Bali and Hite, we group in Panels A and B dividends in \$0.025 intervals except for the exact tick multiples, which are grouped separately. Similarly, in Panels C to F we group dividends in \$0.0125 intervals except for the exact tick multiples, which are again grouped separately.

Panels A and B show that the overnight price drops conform more to the step-function pattern; within ticks, the average price drop increases very slowly with dividend, but right after the tick, there is a jump in the average price drop. The close-to-close price drops show a less clear pattern. This is probably related to the excess noise associated with the close-to-close price drops. In Panels C to F the price drops are grouped in finer intervals, and the within ticks versus between ticks pattern is not as clear as in Panels A and B. According to Fig. 1 the average overnight price drop is often higher than the 45° line. At these points the average price drop is more than the dividend amount. Dividends in the range $0.00 < D \leq 0.125$ have price drops well above zero, and dividends in the range $0.125 < D \leq 0.250$ have price drops considerably larger than \$0.125, in contrast to Bali and Hite's prediction.

In addition to the frequency distributions of ex-day price drops, Table 1 also reports the average price drops for the most common dividend amounts. Bali and Hite predict an average price drop of zero for all dividends less than \$0.125, while Dubofsky predicts that the price drop could be at or slightly above $\bar{D}/2$, which is \$0.0625. For the \$0.05 and \$0.10 dividends the average ex-day price drops are \$0.0739 and \$0.0838, respectively, consistent with Dubofsky. For dividends between \$0.125 and \$0.25 Bali and Hite predict a price drop of \$0.125, whereas Dubofsky predicts the price drop could be somewhat larger than $\bar{D}/2$, which is \$0.125. For the \$0.15, \$0.20, and \$0.25 dividends, the average ex-day price drops are \$0.1828, \$0.1734, and \$0.1914, respectively, again consistent with Dubofsky.

Table 3 presents a regression analysis of the same predictions presented graphically in Fig. 1. Following Bali and Hite (1998), we regress individual price drops on the tick below the dividend ($\underline{D}$) and the discreteness ($\delta = D - \underline{D}$). Panels A–C are for the three subsamples and are using the pooled data, in which each

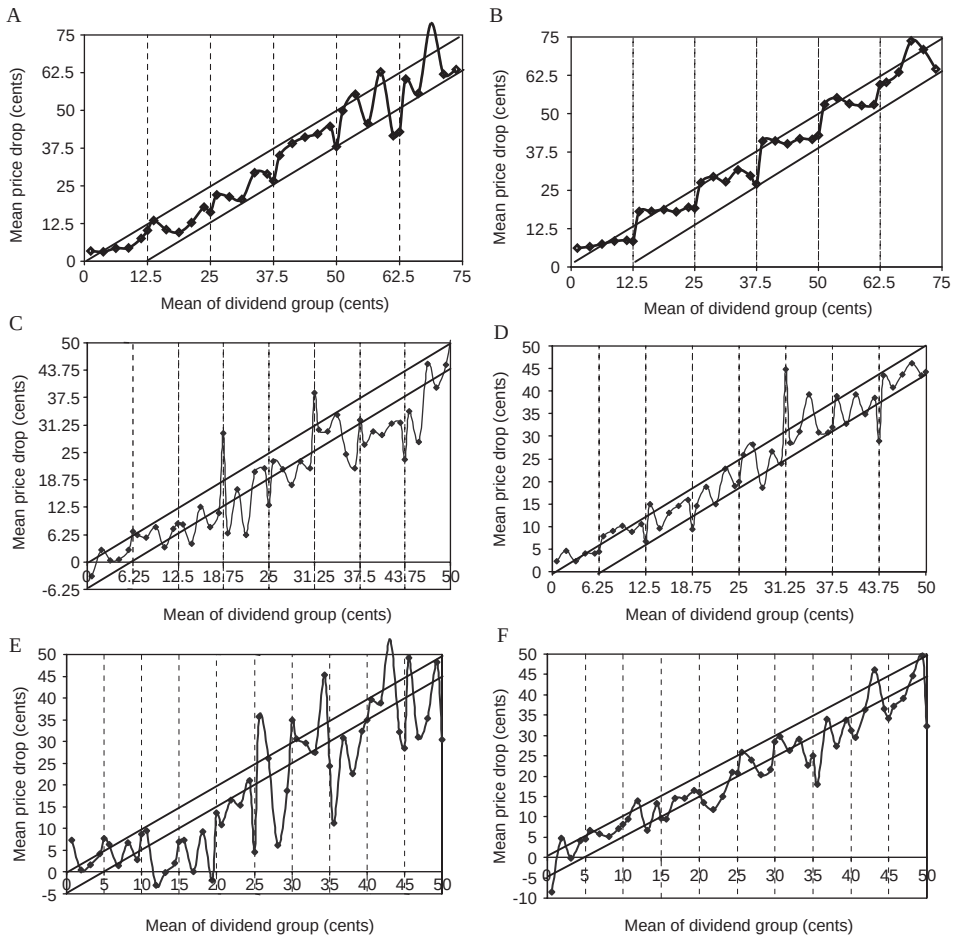


Fig. 1. Mean price drop by dividend groups. The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with cum-day prices of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Panels A and B are for the subsample with a tick size of \$1/8, Panels C and D are for the subsample with a tick size of \$1/16, and Panels E and F are for the subsample with a tick size of \$0.01. In the first two panels, dividends are grouped in two-and-one-half-cent intervals except those dividends that are exact tick multiples, which are grouped separately. In the last four panels, dividends are grouped in one-and-one-quarter-cent intervals except those dividends that are exact tick multiples, which are grouped separately. We then calculate the mean price drop for each dividend group and the results are plotted. Panels A, C, and E are close-to-close price drops; the rest are overnight price drops. The vertical lines indicate exact tick multiples; the two diagonal lines are 45°. One starts at the origin, and the other starts at a tick below the origin.

dividend distribution is treated as a distinct observation. However, dividend distributions tend to cluster in calendar time. To alleviate the effect of ex-date clustering on the regression results, we also present a set of results using a panel data

Table 3

Estimated coefficients for regressions of price drops on dividends and discreteness

This table reports regression results of price drops on the tick below dividend (D) and dividend discreteness ($\delta = D - \underline{D}$), for three subsamples. The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with cum-day prices of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Panels A–C use the pooled data. The model is

$$(\text{price drop})_{i,t} = a + b\underline{D}_{i,t} + b\delta_{i,t} + \varepsilon_{i,t}.$$

In Panels D–F the model is estimated using a panel data approach, in which each ex-day is allowed a fixed effect that affects all price drops on that day. The model is

$$(\text{price drop})_{i,t} = a + \sum_t b_t (\text{fixed effect})_t + c\underline{D}_{i,t} + d\delta_{i,t} + \varepsilon_{i,t}.$$

Panels D–F include only data that contain at least 10 dividend distributions on a given date. Numbers in parentheses are standard errors. For brevity the estimates of the coefficients on the fixed effects are not reported. For the decimalized subsample, we assume the effective tick is \$0.05, because trades tend to occur at five-cent intervals.

Dependent variable	Intercept	Tick below dividend ($\underline{D}$)	Discreteness ($\delta = D - \underline{D}$)	Adjusted R^2	Number of observations
<i>Panel A. Subsample with tick size = \$1/8</i>					
Close-to-close	−0.0062 (0.0078)	0.9385** (0.0195)	0.6481** (0.0981)	0.0874	24,098
Overnight	0.0604** (0.0035)	0.9114** (0.0087)	0.1532** (0.06574)	0.3124	24,098
<i>Panel B. Subsample with tick size = \$1/16</i>					
Close-to-close	−0.0135 (0.01312)	0.8212** (0.0432)	0.6434** (0.3001)	0.0168	21,015
Overnight	0.0208** (0.0065)	0.8900** (0.0212)	0.3116** (0.1475)	0.0776	21,015
<i>Panel C. Subsample with tick size = \$0.01</i>					
Close-to-close	−0.0117 (0.0221)	0.8013** (0.0733)	0.4963 (0.6008)	0.0233	4,941
Overnight	−0.0005 (0.0100)	0.8254** (0.0331)	0.7088** (0.2716)	0.1117	4,941
<i>Panel D. Subsample with tick size = \$1/8</i>					
Close-to-close	0.0028 (0.0084)	0.9106** (0.0207)	0.5516** (0.1013)	0.1347	21,937
Overnight	0.0625** (0.0038)	0.9052** (0.0093)	0.1838** (0.0455)	0.3422	21,937
<i>Panel E. Subsample with tick size = \$1/16</i>					
Close-to-close	−0.0288** (0.0139)	0.8209** (0.0447)	0.7746** (0.3044)	0.0825	19,537
Overnight	0.0116 (0.0067)	0.8937** (0.0217)	0.4273** (0.1474)	0.1255	19,537
<i>Panel F. Subsample with tick size = \$0.01</i>					
Close-to-close	0.0042 (0.0224)	0.8118** (0.0724)	0.0866 (0.5929)	0.1120	4,489

Table 3. (Continued)

Dependent variable	Intercept	Tick below dividend ($\underline{D}$)	Discreteness ($\delta = D - \underline{D}$)	Adjusted R^2	Number of observations
Overnight	0.0025 (0.0105)	0.8146** (0.0337)	0.7695** (0.2760)	0.1511	4,489

**Significantly different from 0 (for the intercept and discreteness) or from 1 (for $\underline{D}$) at the 5% confidence level.

model proposed by Hayashi and Jagannathan (1990) as outlined in Section 3. These results are reported in Panels D to F.⁶ Panels A and D include all dividends that are no greater than \$1.00; other panels include all dividends that are no greater than \$0.50, to reduce the effect of outliers. (Bali and Hite also exclude the cases in which the dividend amount is less than or equal to one tick. We feel no justification exists for this exclusion and hence include these small dividends, which constitute a large part of the sample.) In Panels C and F we take the effective tick size as \$0.05 in constructing $\underline{D}$ and δ .

According to Bali and Hite, the coefficient on $\underline{D}$ should be one, and both the intercept and the coefficient on δ should be zero. According to Table 3, the coefficient on $\underline{D}$ is always different from one at the 5% significance level; and in 10 out of 12 cases the coefficient on δ is significantly different from zero at the 5% confidence level. We also perform the joint test that the coefficient on $\underline{D}$ should be one and both the intercept and the coefficient on δ should be zero. In every case this joint hypothesis is rejected at the 1% confidence level. (These results are not reported in Table 3.) Overall, the evidence is inconsistent with the prediction of Bali and Hite.

4.4. Cum- to ex-day changes in the bid and ask prices and the bid-ask spread

Table 4 contains the summary statistics for the dividends and different measures of changes in market quotes from cum-day to ex-day. We include both bid price and ask price as well as three measures of changes for each type of quote: from cum-day open to ex-day open, from cum-day close to ex-day close, and from cum-day close to ex-day open. Panel A is for the subsample with the tick size of \$1/8; Panel B, the tick size of \$1/16; and Panel C, the tick size of \$0.01. In all three panels the average price drop from cum-day close ask to ex-day open ask is substantially smaller than the average dividend, which in turn is smaller than the average price drop from cum-day close bid to ex-day open bid. The asymmetric changes of the bid and ask quoted prices are consistent with Dubofsky's predictions.

⁶The asymptotic distribution theory of the OLS estimator in the panel data model requires the number of observations in all cross-sections to be large [see Eq. (A.6)]. In Panels D to F of Table 3, we require the cross-sections to have at least ten observations (i.e., an event day to have at least ten dividend distributions). The alternatives of requiring the cross-sections to have at least five or 15 observations give qualitatively similar results. For brevity, the coefficients for the ex-day-specific fixed effects are not reported in the table.

Table 4

Dividends and changes in market quotes for three subsamples

We include changes in market quotes from cum-day open (and close) ask (and bid) to ex-day open (and close) ask (and bid) and from cum-day close ask (and bid) to ex-day open ask (and bid). The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with cum-day prices of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes.

Variable	Mean	Std. Dev.	Minimum	Maximum
<i>Panel A. Subsample with tick size = \$1/8 (24,270 observations)</i>				
Dividend	0.201	0.237	0.0001	20.000
Open ask price drop	0.154	0.587	−12.750	18.000
Open bid price drop	0.170	0.592	−10.375	17.875
Close ask price drop	0.158	0.569	−20.500	17.375
Close bid price drop	0.160	0.575	−23.500	17.250
Price drop from cum-day close ask to ex-day open ask	0.153	0.341	−7.500	20.000
Price drop from cum-day close bid to ex-day open bid	0.215	0.347	−6.375	20.125
<i>Panel B. Subsample with tick size = \$1/16 (22,512 observations)</i>				
Dividend	0.195	0.225	0.0003	12.000
Open ask price drop	0.151	0.887	−18.000	23.000
Open bid price drop	0.167	0.891	−17.500	28.000
Close ask price drop	0.137	0.872	−19.000	16.250
Close bid price drop	0.136	0.892	−27.938	16.188
Price drop from cum-day close ask to ex-day open ask	0.120	0.494	−10.000	12.875
Price drop from cum-day close bid to ex-day open bid	0.205	0.500	−9.875	13.000
<i>Panel C. Subsample with tick size = \$0.01 (5,397 observations)</i>				
Dividend	0.203	0.266	0.001	10.591
Open ask price drop	0.103	0.704	−11.150	10.910
Open bid price drop	0.117	0.712	−11.250	12.000
Close ask price drop	0.138	0.684	−7.820	10.450
Close bid price drop	0.142	0.685	−7.900	10.450
Price drop from cum-day close ask to ex-day open ask	0.118	0.399	−2.750	10.100
Price drop from cum-day close bid to ex-day open bid	0.211	0.412	−2.500	10.800

Table 5 reports the size of the bid-ask spreads in each of the three subsamples. As the tick size is reduced, the bid-ask spread also shrinks. This result is expected, and it is consistent with the literature. Even though the spreads shrink as discreteness is eliminated, some common features still exist in all three subsamples. From cum-day close to ex-day open the bid-ask spread increases in all three subsamples. More important, the ex-day open bid-ask spread is also larger than the cum-day open bid-ask spread in all three subsamples. These findings are consistent with Dubofsky's predictions.

Table 5

The bid-ask spreads on cum- and ex-days at market open and close for three subsamples

The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with a cum-day price of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes.

Variable	Mean	Std. Dev.	Minimum	Maximum
<i>Panel A. Subsample with tick size = \$1/8 (24,270 observations)</i>				
Cum-day open bid-ask spread	0.242	0.156	0.125	10.000
Cum-day close bid-ask spread	0.196	0.102	0.125	4.000
Ex-day open bid-ask spread	0.258	0.162	0.000	6.750
Ex-day close bid-ask spread	0.198	0.101	0.125	3.250
<i>Panel B. Subsample with tick size = \$1/16 (22,512 observations)</i>				
Cum-day open bid-ask spread	0.214	0.310	0.0625	22.000
Cum-day close bid-ask spread	0.145	0.279	0.0625	30.000
Ex-day open bid-ask spread	0.230	0.386	0.0000	36.000
Ex-day close bid-ask spread	0.143	0.133	0.0625	10.000
<i>Panel C. Subsample with tick size = \$0.01 (5,397 observations)</i>				
Cum-day open bid-ask spread	0.168	0.186	0.010	5.100
Cum-day close bid-ask spread	0.089	0.112	0.010	2.300
Ex-day open bid-ask spread	0.182	0.189	0.010	4.000
Ex-day close bid-ask spread	0.093	0.107	0.010	2.010

Table 6 presents statistical analyses of the observed changes in bid-ask spreads from cum- to ex-dividend day at both the market open and the market close. Panels A to C treat each dividend distribution as a distinct observation. To avoid the ex-date clustering problem, Panels D–F are estimates of the panel data model described in Section 3. (In estimating the panel data model, we require an ex-day to have at least ten observations to be included in the sample. Reducing the required observations per day to five or increasing the number to 15 does not qualitatively change the result.) Table 6 shows strong evidence that the open spread is significantly wider on the ex-day than on the cum-day. These results again are consistent with the idea that the bid-prices drop more than the ask prices from cum-day to ex-day, as predicted by Dubofsky.

4.5. Overall impact of decimalization on the price drop anomaly

Bali and Hite's model predicts that the price drop anomaly should be reduced with reductions in discreteness. However, Dubofsky's model predicts that the anomaly could remain or even exacerbate. Table 7 contains the summary statistics for the dividends and different measures of ex-dividend day price drops. We include three measures of price drops: from cum-day open to ex-day open, cum-day close to ex-day close, and cum-day close to ex-day open. Panel A of Table 7 is the result for the subsample with the tick size of \$1/8, Panel B is for the subsample with the tick size of \$1/16, and Panel C is for the subsample with the tick size of one cent. Across the

Table 6

Changes in bid-ask spreads from cum- to ex-dividend day at market open and at market close

The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with a cum-day price of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Panels A–C treat each dividend distribution as a distinct observation. Numbers in Panels D–F are the estimates of a and its standard error (in parentheses) in the panel data model

$$(\text{ex-day spread} - \text{cum-day spread})_{i,t} = a + \sum_t b_t (\text{fixed effect})_t + \varepsilon_{i,t}.$$

We require an ex-day to have at least ten dividend distributions to be included in the sample that is used to estimate the panel data model.

Changes in spread	Number of observations	Mean
<i>Panel A. Subsample with tick size = \$1/8</i>		
Open spread	24,270	0.0160**
Close spread	24,270	0.0022**
<i>Panel B. Subsample with tick size = \$1/16</i>		
Open spread	22,512	0.0156**
Close spread	22,512	−0.0015
<i>Panel C. Subsample with tick size = \$0.01</i>		
Open spread	5,397	0.0138**
Close spread	5,397	0.0038*
<i>Panel D. Subsample with tick size = \$1/8</i>		
Open spread	24,098	0.0160** (0.0013)
Close spread	24,098	0.0029** (0.0008)
<i>Panel E. Subsample with tick size = \$1/16</i>		
Open spread	21,014	0.0138** (0.0016)
Close spread	21,014	0.00086 (0.0010)
<i>Panel F. Subsample with tick size = \$0.01</i>		
Open spread	4,940	0.0125** (0.0035)
Close spread	4,940	0.0026 (0.0017)

* Significant at the 5% confidence level.

** Significant at the 1% confidence level.

three subsamples, the average dividends are very similar. The average dividends over the three subsamples are \$0.201, \$0.195, and \$0.203, respectively. The average close-to-close price drops are \$0.164, \$0.138, and \$0.142, respectively. The average cum-day close to ex-day open price drops are \$0.199, \$0.173, and \$0.165, respectively. As the tick size declined, the average difference between dividend and ex-day price drop

Table 7

Dividends and price drops

We include three measures of ex-dividend day price drop: cum-day close price to ex-day close price, cum-day open price to ex-day open price, and cum-day close to ex-day open price. All variables are measured in dollars. The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with cum-day prices of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Observations that have multiple dividend distributions by the same firm on the same day are excluded.

Variable	Mean	Std. Dev.	Minimum	Maximum
<i>Panel A. Subsample with tick size = \$1/8 (24,270 observations)</i>				
Cum-day close price	27.175	22.697	2.000	702.750
Dividend	0.201	0.237	0.0001	20.000
The tick below dividend ($\underline{D}$)	0.136	0.236	0	19.875
Close price drop	0.164	0.575	−22.000	17.500
Open price drop	0.171	0.563	−9.750	18.000
Price drop from cum-day close to ex-day open	0.199	0.325	−4.500	20.250
Dividend – close price drop	0.0375	0.533	−10.800	22.050
Dividend – cum-close to ex-open price drop	0.0030	0.234	−4.700	4.550
Ex-day open to close price change	0.0351	0.477	−7.875	15.500
<i>Panel B. Subsample with tick size = \$1/16 (22,512 observations)</i>				
Cum-day close price	27.975	27.363	2.000	600.438
Dividend	0.195	0.225	0.0003	12.000
The tick below dividend ($\underline{D}$)	0.162	0.226	0	11.9375
Close price drop	0.138	0.872	−19.375	16.250
Open price drop	0.164	0.862	−18.000	22.000
Price drop from cum-day close to ex-day open	0.173	0.467	−10.938	12.938
Dividend – close price drop	0.0569	0.851	−15.750	19.595
Dividend – cum-close to ex-open price drop	0.0236	0.424	−12.670	12.220
Ex-day open to close price change	0.0353	0.734	−15.500	13.375
<i>Panel C. Subsample with tick size = \$0.01 (5,397 observations)</i>				
Cum-day close price	26.132	22.219	2.000	597.260
Dividend	0.203	0.266	0.001	10.591
The tick below dividend ($\underline{D}$)	0.174	0.266	0	10.550
Close price drop	0.142	0.685	−7.860	10.460
Open price drop	0.114	0.678	−5.150	10.900
Price drop from cum-day close to ex-day open	0.165	0.388	−2.800	10.500
Dividend – close price drop	0.0609	0.646	−7.350	8.030
Dividend – cum-close to ex-open price drop	0.0389	0.307	−3.920	4.255
Ex-day open to close price change	0.0227	0.565	−6.030	7.700

Table 8

The average difference between dividends and price drops

The sample period is January 1993 to December 2001. We include all ordinary cash taxable distributions to NYSE stocks with a cum-day price of \$2.00 or more and with available cum-day and ex-day open and close trade prices and quotes. Numbers in the table are the estimate of a and its standard error (in parentheses) in the panel data model

$$(\text{dividend} - \text{price drop})_{i,t} = a + \sum_i b_i (\text{fixed effect})_i + \varepsilon_{i,t}.$$

We require an ex-day to have at least 10 dividend distributions to be included in the sample that is used to estimate the panel data model.

Variable	Mean
<i>Panel A. Subsample with tick size = \$1/8 (24,098 observations)</i>	
Dividend – close price drop	0.0380** (0.0041)
Dividend–cum-close to ex-open price drop	0.0032* (0.0018)
<i>Panel B. Subsample with tick size = \$1/16 (21,014 observations)</i>	
Dividend – close price drop	0.0583** (0.0068)
Dividend – cum-close to ex-open price drop	0.0208** (0.0034)
<i>Panel C. Subsample with tick size = \$0.01 (4,941 observations)</i>	
Dividend – close price drop	0.0457** (0.0108)
Dividend – cum-close to ex-open price drop	0.0266** (0.0050)

* Significantly different from zero at the 10% confidence level.

** Significantly different from zero at the 5% confidence level.

increased somewhat, whether we measure price drop from close to close or from cum-day close to ex-day open.

We do not control for the ex-day clustering problem in Table 7. We use the panel data model to control for the clustering problem and estimate the average difference between dividends and ex-day price drops. These results are reported in Table 8. (In estimating the panel data model, we require an ex-day to have at least ten observations to be included in the sample. Reducing the required observations per day to five or increasing the number to 15 does not qualitatively change the result.) Again, the average price drops are significantly smaller than the average dividends in all cases. No evidence exists that, as tick size declined, the average difference between dividend and ex-day price drop decreased, whether we measure price drop from close to close or from cum-day close to ex-day open.

5. Concluding remarks

Dubofsky (1992) and Bali and Hite (1998) offer novel market microstructure explanations to the ex-dividend day price drop anomaly. Bali and Hite's model

predicts that the ex-day price drop should equal the tick below the dividend. Dubofsky's model suggests that both the mechanics of NYSE Rule 118 and price discreteness are involved in the price drop anomaly. Using a relatively large sample of data with different tick sizes, we examine the predictions of both models. Our evidence is inconsistent with Bali and Hite's prediction and qualitatively consistent with Dubofsky's prediction. But if we interpret Dubofsky's fully constrained model literally, it is obviously rejected by the data.

Finally, both Bali and Hite and Dubofsky are correct in recognizing that tick size is an important factor in the ex-day price drop anomaly. However, even though we find no inconsistency with Dubofsky at a qualitative level, at this level his model gives no predictions about the exact amount of the ex-day price drop. Hence the model is not a complete explanation of the ex-day price drop anomaly.

Appendix A. The fixed-effect panel data model

We discuss the panel data modeling of Eq. (5) only. The treatment of Eq. (7) is similar. Allowing a fixed effect for each ex-day, Eq. (5) becomes

$$\Delta P_{t,i} = a_t + b\underline{D}_{t,i} + c\delta_{t,i} + \varepsilon_{t,i}. \quad (\text{A.1})$$

Grouping together all the relevant variables that occur on the same ex-date into column vectors, and using bold symbols to denote them, we can rewrite Eq. (A.1) as

$$\Delta \mathbf{P}_t = a_t \mathbf{1} + b\underline{\mathbf{D}}_t + c\delta_t + \boldsymbol{\varepsilon}_t, \quad t = 1, 2, \dots, T, \quad (\text{A.2})$$

where $\mathbf{1}$ is the column vector of ones and all the vectors have length N_t . We can further stack the column vectors in Eq. (A.2) together and rewrite it as

$$\Delta \mathbf{P} = \sum_{t=1}^T a_t \mathbf{e}_t + b\underline{\mathbf{D}} + c\delta + \boldsymbol{\varepsilon}, \quad (\text{A.3})$$

where the vectors are all of length $(\sum_{t=1}^T N_t)$ and $\mathbf{e}_t$ has N_t ones corresponding to the observations that fall on date t , and zero everywhere else.

To isolate the common intercept term in Eq. (A.3), which is of interest to us, we rewrite Eq. (A.3) again as

$$\Delta \mathbf{P} = \bar{a} \mathbf{1} + \sum_{t=1}^T (a_t - \bar{a}) \mathbf{e}_t + b\underline{\mathbf{D}} + c\delta + \boldsymbol{\varepsilon}, \quad (\text{A.4})$$

where $\bar{a}$ is the average of a_t and $\mathbf{1}$ is again the vector of ones. Because $\sum_{t=1}^T (a_t - \bar{a}) = 0$, we can rewrite Eq. (A.4) as

$$\Delta \mathbf{P} = \bar{a} \mathbf{1} + \sum_{t=1}^{T-1} a'_t (\mathbf{e}_t - \mathbf{e}_T) + b\underline{\mathbf{D}} + c\delta + \boldsymbol{\varepsilon}. \quad (\text{A.5})$$

We will apply the OLS procedure to estimate the coefficients $\bar{a}$, a'_t , b , and c , even though we are interested in the coefficients $\bar{a}$, b , and c only. We will assume that T is

fixed and

$$\min_t (N_t) \rightarrow +\infty. \quad (\text{A.6})$$

Then the standard asymptotic distribution theory of the OLS estimator applies (see [Greene, 1993, pp. 295–297](#)).

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